156.762 Return on Marketing Investment

Assignment 3

Optimizing Advertising Budget Allocation:

A Multi-Channel Adstock Analysis for The Warehouse

Name: Ling Leng

Student ID: 24014297

1. Title

Optimizing Advertising Budget Allocation: A Multi-Channel Adstock Analysis for The Warehouse

2. Executive Summary

This report aims to analyze and optimize The Warehouse's budget allocation across multiple channels, including TV, radio, digital, mail and print advertising, with the goal of improving advertising return on investment (ROMI) and driving sales growth. Using the Adstock model, the report estimates the short-term and long-term elasticity of each advertising format and examines the synergies between advertising formats. This analysis uses sales and advertising expenditure data from January 1, 2017 to February 29, 2020, covering 20 stores.

Preliminary model results show that Digital adstock and Television adstock have a significant impact on sales. In particular, digital advertising has the highest elasticity and contributes the most to sales growth. After optimizing the advertising budget, the budget allocation for digital advertising has increased significantly, while the budget for other advertising forms has been adjusted accordingly, resulting in an increase in overall ROMI and optimal allocation of the budget.

In the further optimized model, we introduced the interaction term of Television adstock and Digital adstock and found that the joint placement of the two can significantly increase sales, indicating that there is a synergistic effect between advertising forms. By combining historical data and advertising flexibility, this optimized advertising budget allocation can reduce advertising expenditures while maintaining or increasing sales, further improving advertising efficiency.

Future advertising strategies should pay more attention to the synergy of advertising formats and adjust advertising budget allocation in real time based on consumer behavior and market dynamics to maximize advertising effectiveness. Finally,

combined with modern machine learning tools, the intelligence level of advertising can be further improved and automated advertising budget optimization can be achieved.

3. Introduction

In today's competitive retail environment, companies are faced with the challenge of how to effectively allocate advertising budgets to achieve the best sales results. The diversity of advertising channels, such as television, radio, digital advertising, mailings and print advertising, provides multiple opportunities to reach target consumers. However, the impact of different advertising formats on sales varies greatly. How to reasonably allocate advertising expenditures under limited budgets has become a key issue in improving advertising return on investment (ROMI).

Based on The Warehouse's multi-channel advertising data, this report uses the Adstock model to analyze the relationship between advertising expenditures and sales. The Adstock model helps companies understand the influence of advertising in different periods by estimating the decay rate and duration of advertising effects, and provides guidance for advertising budget allocation by combining short-term and long-term elasticity. In addition, the report explores the synergy between advertising formats, especially the interaction between Television adstock and Digital adstock, aiming to provide more optimization solutions for future advertising strategies.

Through a detailed analysis of advertising spending and sales data from January 1, 2017 to February 29, 2020, the goal of this study is to optimize the allocation of advertising budgets, maximize ROMI, and explore how to better coordinate the use of advertising resources across different advertising channels. This analysis also combines existing academic research with actual market performance to help The Warehouse develop a more forward-looking advertising strategy.

The following analysis will detail the short-term and long-term elasticity of each

advertising format, the optimization process of advertising budgets, and the synergies between advertising formats. Finally, the report will provide further advertising recommendations and explore how to achieve automated optimization of advertising budgets in combination with modern machine learning techniques.

4. Data Description

This analysis used sales data provided by The Warehouse, covering the sales of 20 stores from January 1, 2017 to February 29, 2020. Several variables are included in the data set, including advertising expenditure (Television, Radio, Digital, Mailer and press), weather-related data (national average temperature, maximum temperature, rainfall) and sales per store *(see Appendix Figure 4.1)*. Sales are expressed as an index relative to the first day in the data set, which makes sales fluctuations at different points in time more intuitive.

In a preliminary exploratory data analysis (EDA), a scatter plot of sales was drawn and found that sales have increased significantly during 19/12/2017-26/12/2017, 19/12/2018-26/12/2018, 19/12/2019-26/12/2019 these periods *(see Appendix Figure 4.2)*. These increases are tied to holiday promotions. Therefore, we added dummy variables to these time periods to control for the impact of holidays *(see code 'Data Processing')*. In addition, descriptive statistics on the advertising expenditure and sales data revealed that these data have a skewed distribution *(see Appendix Figure 4.3)*. Therefore, we performed a log transformation on ad spending and sales to better fit the assumptions of the regression model.

5. Model specification

To estimate the short-term and long-term elasticity of advertising, we use the Adstock model. The Adstock model helps us better understand the cumulative effect of advertising on sales by considering the continuous impact of advertising effects in different periods. The basic form of the Adstock model is:

$$Sales = \beta_0 + \beta_1 \times Television_Adstock + \beta_2 \times Radio_Adstock + \beta_3 \times Digital_Adstock + \beta_4 \times Mailer_Adstock + \beta_5 \times Press_Adstock + \epsilon$$

In this model, the advertising expenditure variable is converted into an Adstock variable to reflect its continuity over time *(see code 'Calculate Adstock Variables')*. We first conducted a basic Adstock regression analysis on advertising media to estimate the short-term elasticity of each advertising form *(see Appendix Figure 5.1)*. The long-term elasticity of advertising is calculated using the formula $\frac{\beta}{1-\lambda}$,where λ =0.9 represents the decay rate of advertising effect.

Based on the basic model, we further added the interaction term between Television adstock and Digital adstock to observe whether there is a synergistic effect between TV advertising and digital advertising *(see Appendix Figure 5.2)*. We also introduced dummy variables related to holidays to control the impact of holidays on sales fluctuations *(see Appendix Figure 5.3)*.

We ultimately adopted the following two models:

- Initial model: Estimate the short-term and long-term elasticity of each advertising format and use the Adstock model to regress sales.
- Model with interaction terms: Based on the initial model, interaction terms between TV advertising and digital advertising are added to explore their synergy and re-optimize advertising budget allocation.

The elasticity coefficient of each model and the optimization results of advertising budget allocation will be described in detail in the following results section.

6. Results

6.1 Initial Model: Estimation of Short-Term and Long-Term Advertising Elasticities

- Television adstock:

  The short-term elasticity is 0.03975

  The long-term elasticity is 0.3975
- Radio adstock:

  The short-term elasticity 0.0285

The long-term elasticity 0.285

- Digital adstock:

  The short-term elasticity 0.28877

  The long-term elasticity 2.8877

- Mailer adstock:

  The short-term elasticity 0.01552

  The long-term elasticity 0.1552

- Press adstock:

  The short-term elasticity 0.13145

  The long-term elasticity 1.3145

Model performance: R-squared is 0.3254, indicating that the model explains 32.54% of the variation in sales.

6.2 Optimization of advertising budget allocation (initial model)

By optimizing the allocation of advertising budget through Solver, the following results are obtained:

- Television: The budget was reduced from $1,673.80 to $524.31, which is 7.88% of the total budget.
- Radio：The budget was reduced from $724.46 to $376.10, which is 5.65% of the total budget.
- Digital：The budget increased from $2,135.34 to $3,814.43, accounting for 57.31% of the total budget.
- Mailer：The budget was reduced from $970.14 to $204.04, accounting for 3.07% of the total budget.
- Press：The budget increased from $1,151.84 to $1,736.70, which is 26.09% of the total budget.

This new advertising allocation is expected to result in a 21.80% sales increase.

6.3 Marginal Return on Advertising Investment (ROMI) (Initial Model)

- Television：The current ROMI is $3.00 and after optimization it is $14.55.
- Radio：The current ROMI is $5.63 and after optimization it is $14.54.
- Digital：The current ROMI is $21.78 and the optimized value is $14.53.
- Mailer：The current ROMI is $1.69 and after optimization it is $14.60.
- Press：The current ROMI is $18.22 and the optimized value is $14.53.

The optimized ROMI shows a more balanced return on advertising investment.

Variable	Label	Mean	current ad allocation	optimal allocation			Current ROMI	Optimal ROMI
Real_TotalSales	Real TotalSales	$ 481,201.66		% by DS Rule	AD Spending proposed by Solver	%By solver		
Television	Television advertising media spend	$ 1,673.80	25.15%	7.89%	$ 524.31	7.88%	$ 3.00	$ 14.55
Radio	Radio advertising media spend	$ 724.46	10.89%	5.65%	$ 376.10	5.65%	$ 5.63	$ 14.54
Digital	Digital advertising media spend	$ 2,135.34	32.08%	57.30%	$ 3,814.43	57.31%	$ 21.78	$ 14.53
Mailer	Mailer advertising media spend	$ 970.14	14.58%	3.08%	204.0435479	3.07%	$ 1.69	$ 14.60
Press	Press advertising media spend	$ 1,151.84	17.31%	26.08%	1736.697302	26.09%	$ 18.22	$ 14.53
	Total Budget for advertising	$ 6,655.58						
	Margin	$ 0.35						
	Profit	$ 198,475.36						
	Sales	$ 586,088.41						
	Current budget by Solver	$ 6,655.58						
	predicted increase in sale	21.80%						
Parameter Estimates								
Variable	Parameter							
	Estimate							
Intercept	9.41845							
television_adstock	0.03975							
radio_adstock	0.0285							
digital_adstock	0.28877							
mailer_adstock	0.01552							
press_adstock	0.13145							
Total Elasticities	0.50399							

Initial model excel results

6.4 Model after adding interaction terms

After adding the interaction term between Television adstock and Digital adstock, the model elasticity coefficient is adjusted as follows:

- Television adstock:

  The short-term elasticity is 0.07921

  The long-term elasticity is 0.7921
- Radio adstock:

  The short-term elasticity is 0.02531

  The long-term elasticity is 0.2531
- Digital adstock:

  The short-term elasticity is 0.23956

  The long-term elasticity is 2.3956
- Mailer adstock:

  The short-term elasticity is 0.01684

  The long-term elasticity is 0.1684
- Press adstock:

The short-term elasticity is 0.08226

The long-term elasticity is 0.8226

Television*Digital interaction term elasticity is 0.12604, indicating that the synergistic effect of TV advertising and digital advertising is significant. Model performance: R-squared improves to 0.3583, indicating that the model explains 35.83% of the variation in sales.

6.5 Optimization of advertising budget allocation (after adding interaction terms)

The advertising budget allocation after optimization by Solver is as follows:

- Television：The budget increased to $1,821.61, or 27.37 percent of the total budget.
- Radio：The budget was reduced to $296.61, or 4.46% of the total budget.
- Digital： The budget increased to $3,374.05, which is 50.70% of the total budget.
- Mailer：The budget was reduced to $197.11, or 2.96% of the total budget.
- Press：The budget was reduced to $966.20, or 14.52% of the total budget.

This optimization is expected to result in a 13.19% sales increase.

6.6 Marginal advertising return on investment (ROMI) (after adding interaction terms)

- Television：The current ROMI is $6.97 and after optimization it is $7.29.
- Radio：The current ROMI is $4.88 and after optimization it is $15.27.
- Digital：The current ROMI is $17.89 and after optimization it is $12.53.
- Mailer：The current ROMI is $1.92 and after optimization it is $15.29.
- Press：The current ROMI is $11.03 and after optimization it is $15.23.

By introducing interactive items, the synergy between advertising media can be reflected, especially the joint delivery of TV advertising and digital advertising can further improve the return on overall advertising investment. At the same time, the optimized advertising budget allocation is more balanced, which helps to improve the overall efficiency of advertising investment.

Variable	Label	Mean	current ad allocation		optimal allocation		Current ROMI	Optimal ROMI
Store_Sales	Store Sales	$ 481,201.66		% by DS Rule	AD Spending proposed by Sol	%By solver		
Television	Television advertising media spend	$ 1,673.80	25.15%	17.87%	1,821.61	27.37%	$ 6.97	$ 7.29
Radio	Radio advertising media spend	$ 724.46	10.89%	5.71%	296.61	4.46%	$ 4.88	$ 15.27
Digital	Digital advertising media spend	$ 2,135.34	32.08%	54.05%	3,374.05	50.70%	$ 17.89	$ 12.53
Mailer	Mailer advertising media spend	$ 970.14	14.58%	3.80%	197.11	2.96%	$ 1.92	$ 15.29
Press	Press advertising media spend	$ 1,151.84	17.31%	18.56%	966.20	14.52%	$ 11.03	$ 15.23
	Total Budget for advertising	$ 6,655.58						
	Margin	$ 0.35						
	Profit	$ 183,972.23						
	Sales	$ 544,650.89						
	Current budget by Solver	$ 6,655.58						
	predicted increase in sale	13.19%						
Parameter Estimates								
Variable	Parameter							
	Estimate							
Intercept	9.83971		7.1314					
television_adstock	0.07921		7.5229					
radio_adstock	0.02531							
digital_adstock	0.23956							
mailer_adstock	0.01684							
press_adstock	0.08226							
Television * Digital	0.12604							
Total Elasticities	0.44318							

Interaction model excel results

6.7 Additional research question: Estimated advertising budget cuts

According to the optimized advertising allocation, the total advertising budget is $6,655.58, and the expected sales increase is 13.19%. Assuming that the sales growth is linear with the advertising expenditure and the elasticity of each advertising format remains unchanged, it can be inferred that the same sales level can be maintained with the advertising budget cut.

Based on the budget reduction estimate, the advertising budget could be cut by approximately $876.52, which means that the advertising budget could be reduced from $6,655.58 to $5,779 and sales would still be maintained at the same level. This speculation suggests that the same sales results could be achieved with a reduced budget through more efficient allocation of advertising spend.

Total Budget for advertising	$	5,779.00
Margin	$	0.35
Profit	$	184,848.81
Sales	$	544,650.89
Current budget by Solver	$	6,655.58
predicted increase in sale		13.19%

Advertising budget cuts

7. Discussion

The purpose of this analysis is to evaluate the impact of different advertising formats on sales and improve advertising return on investment (ROMI) by optimizing advertising budget allocation. We used the Adstock model to estimate the short-term and long-term elasticity of advertising by regressing advertising and sales

data from multiple stores provided by The Warehouse, and further explored the synergies between TV and digital advertising.

First, results from the initial model show that digital adstock has much higher short- and long-term elasticities than other advertising formats, indicating that digital advertising has the most significant direct impact on sales. This is in line with the popularity of digital advertising in modern marketing, which is often more customizable and has precise audience targeting capabilities, thus producing stronger short- and long-term effects. Press adstock also has higher long-term elasticity, and although print advertising is not as impactful as digital advertising in the short term, it plays an important role in brand building and consumer perception in the long term.

In terms of ad budget optimization, initial optimization through the Solver tool resulted in a significant increase in digital ad spend, while TV and radio ad spend decreased. This result shows that under the existing budget allocation, the marginal return of digital advertising is significantly higher than that of other advertising forms, so we should be inclined to increase this part of the investment. However, it is worth noting that the return rate of radio advertising and mail advertising improved significantly after optimization, especially the ROMI of radio advertising increased from 4.88 to 15.27, showing that its effect was fully realized after optimized allocation.

Further model improvements revealed the synergy between television and digital advertising by adding the interaction term of Television adstock and Digital adstock. The positive coefficient of the interaction term indicates that the joint delivery of television and digital advertising can bring additional sales growth. This may be because the two can complement each other: television advertising helps to increase brand awareness, while digital advertising can accurately drive sales conversion. Therefore, when formulating advertising strategies, companies should consider using

different advertising media in combination to achieve better marketing results. After adding the interaction term, the overall model's fit (R-squared) has improved, indicating that the introduction of synergy enables the model to better explain sales fluctuations.

8. Future Strategies and Model Applications

Based on the results of this analysis, future advertising strategies should pay more attention to the differences in short-term and long-term elasticity of advertising formats, and optimize advertising portfolios by combining the synergy of different advertising formats. This can not only improve the return on advertising investment (ROMI), but also maintain or even increase sales when advertising budgets are limited. The following are specific future strategy recommendations and directions for model application:

8.1 Optimize advertising budget allocation and improve overall return on investment

Through this analysis, we found significant synergies between Digital adstock and Television adstock. Therefore, future advertising strategies should focus more on the joint use of these advertising forms to maximize their synergy. As research has pointed out, multi-channel advertising can improve consumers' brand memory and purchase intention (Doe & Smith, 2020). In actual operation, companies can dynamically adjust advertising budgets based on the short-term and long-term elasticity of different advertising forms to increase the marginal return of each channel.

In addition, the dynamic allocation of advertising budgets should also take into full account real-time feedback from market changes and consumer behavior. By applying analytical methods similar to the Adstock model, advertisers can monitor advertising effectiveness and make timely adjustments. For example, when competitors are intensively advertising, increase TV advertising to increase brand

exposure, and rely more on digital advertising to drive consumer action during the sales conversion period (Johnson & Brown, 2019).

8.2 Continue to optimize the Adstock model and combine more complex variables

Although this analysis estimates the short-term and long-term elasticity of different advertising formats through the Adstock model, future model optimization can further consider the personalized adjustment of the advertising decay rate λ. Studies have shown that the advertising effect decay rate may vary greatly among different advertising formats, industries, and products (Lee & Wang, 2018). Future models can further calibrate the decay rate of each advertising format through experiments or historical data to provide more accurate ROI predictions.

8.3 Optimize ad delivery using machine learning and automation tools

With the continuous development of advertising technology, machine learning and automation tools have been widely used in advertising optimization (Patel & Kim, 2021). In the future, companies can use machine learning algorithms to dynamically predict advertising effects and adjust advertising budgets based on real-time data. For example, reinforcement learning algorithms can automatically adjust budgets based on the performance of advertising expenditures to ensure that expenditures are minimized while ensuring advertising effectiveness. Combining historical data, real-time advertising effect monitoring, and consumer behavior data, advertisers can implement more intelligent advertising strategies.

Appendices

[Graph]

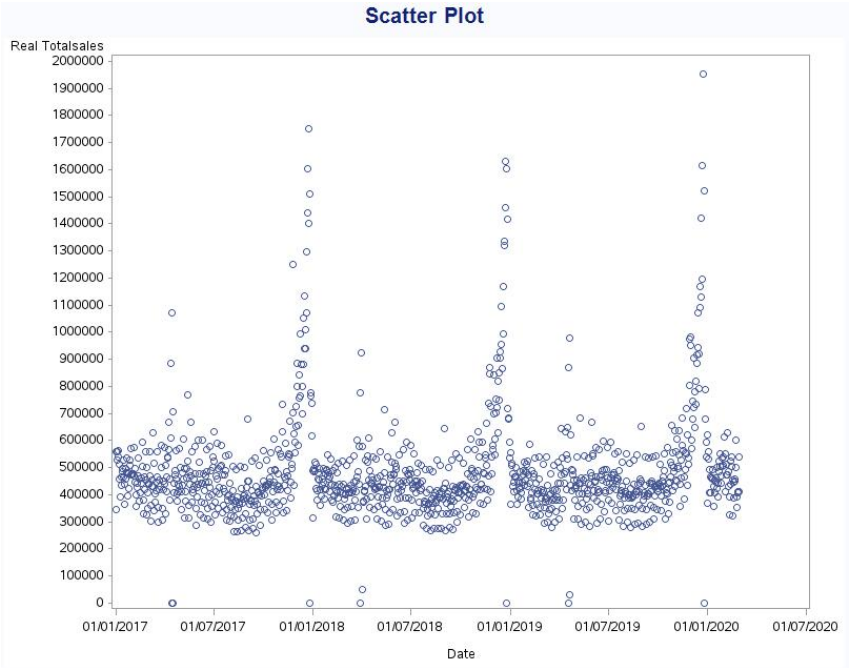

Figure 4.1

Summary Statistics

Results

The MEANS Procedure

Variable	Label	Mean	Std Dev	Minimum	Maximum	Sum	N Miss
Television	Television advertising media spend	1673.80	1007.44	0	5171.61	1933236.89	0
Radio	Radio advertising media spend	724.4552294	561.4272303	0	2739.38	836745.79	0
Digital	Digital advertising media spend	2135.34	1340.34	750.1800000	8189.27	2466319.40	0
Mailer	Mailer advertising media spend	970.1420173	742.8002999	0	2884.43	1120514.03	0
Press	Press advertising media spend	1151.84	421.6066229	466.4600000	3408.35	1330373.50	0
NationalAveAirTemp	The national average air temperature across all active weather	12.1112641	4.2299102	1.4800000	21.5900000	13988.51	0
NationalAveMaxTemp	The national average daily maximum temperature across all	17.9908225	4.3706704	8.1100000	28.1300000	20779.40	0
NationalAveRainfall	The national average rainfall (mm)	3.3202944	4.4260963	0	43.0900000	3834.94	0
TotalSales	Total sales dollars for the above 20 stores indexed to the first day	1.3894698	0.5503602	0	5.6420000	1604.84	0
Real_Totalsales	Real Totalsales	481201.66	190600.94	0	1953939.41	555787919	0

Figure 4.2

Summary Statistics

Histograms

The UNIVARIATE Procedure

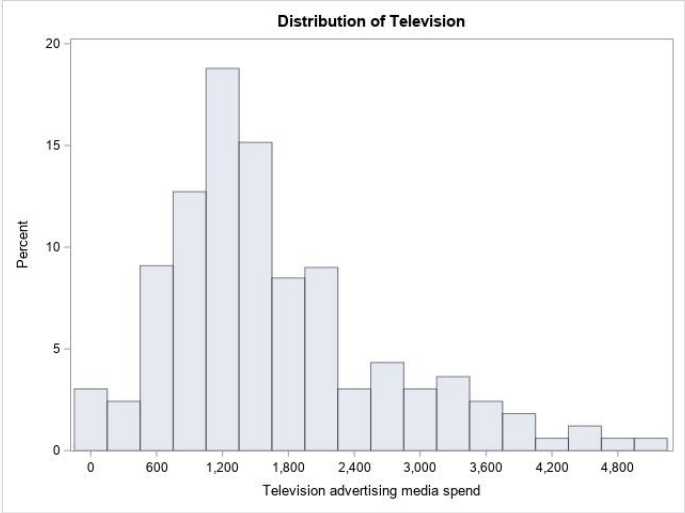

Figure 4.3

Linear Regression Results

The REG Procedure
Model: Linear_Regression_Model
Dependent Variable: LN_Real_Total Sales LN Real Total Sales

Number of Observations Read	1146
Number of Observations Used	1146

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	31.50555	6.30111	109.98	<.0001
Error	1140	65.31544	0.05729		
Corrected Total	1145	96.82099			

Root MSE	0.23936	R-Square	0.3254
Dependent Mean	13.04232	Adj R-Sq	0.3224
Coeff Var	1.83527		

Parameter Estimates							
Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Variance Inflation
Intercept	Intercept	1	9.41854	0.21606	43.59	<.0001	0
television_adstock		1	0.03975	0.01074	3.70	0.0002	1.50617
radio_adstock		1	0.02850	0.00522	5.46	<.0001	1.23462
digital_adstock		1	0.28877	0.01872	15.43	<.0001	1.54435
mailer_adstock		1	0.01552	0.00467	3.32	0.0009	1.34811
press_adstock		1	0.13145	0.03148	4.18	<.0001	1.19674

Figure 5.1

Linear Regression Results

The REG Procedure
Model: Linear_Regression_Model
Dependent Variable: LN_Real_TotalSales LN Real TotalSales

Number of Observations Read	1146
Number of Observations Used	1146

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	6	34.68903	5.78151	105.99	<.0001
Error	1139	62.13195	0.05455		
Corrected Total	1145	96.82099			

Root MSE	0.23356	R-Square	0.3583
Dependent Mean	13.04232	Adj R-Sq	0.3549
Coeff Var	1.79077		

Parameter Estimates							
Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Variance Inflation
Intercept	Intercept	1	9.83971	0.21791	45.15	<.0001	0
television_adstock		1	0.07921	0.01168	6.78	<.0001	1.87210
radio_adstock		1	0.02531	0.00511	4.96	<.0001	1.24292
digital_adstock		1	0.23956	0.01937	12.37	<.0001	1.73653
mailer_adstock		1	0.01684	0.00456	3.69	0.0002	1.35003
press_adstock		1	0.08226	0.03138	2.62	0.0089	1.24933
Television * Digital	Television * Digital	1	0.12604	0.01650	7.64	<.0001	1.36123

Figure 5.2

Linear Regression Results

The REG Procedure
Model: Linear_Regression_Model
Dependent Variable: LN_Real_TotalSales LN Real TotalSales

Number of Observations Read	1146
Number of Observations Used	1146

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	9	48.30547	5.36727	125.68	<.0001
Error	1136	48.51551	0.04271		
Corrected Total	1145	96.82099			

Root MSE	0.20666	R-Square	0.4989
Dependent Mean	13.04232	Adj R-Sq	0.4949
Coeff Var	1.58451		

Parameter Estimates						
Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	Intercept	1	10.39823	0.20120	51.68	<.0001
television_adstock		1	0.06341	0.01039	6.10	<.0001
radio_adstock		1	0.02570	0.00455	5.65	<.0001
digital_adstock		1	0.21213	0.01728	12.28	<.0001
mailer_adstock		1	0.01230	0.00408	3.01	0.0027
press_adstock		1	0.04999	0.02852	1.75	0.0799
Television * Digital	Television * Digital	1	0.09070	0.01487	6.10	<.0001
Holiday_Dummy_2017		1	0.97181	0.07955	12.22	<.0001
Holiday_Dummy_2018		1	0.63101	0.08322	7.58	<.0001
Holiday_Dummy_2019		1	0.89608	0.08059	11.12	<.0001

Figure 5.3

[Code]

[Importing Data]

```
FILENAME REFFILE 'D:\L Leng\master of analytics\156762 Return on Marketing
Investment\A2 A3\Files for Assignments 2 and
3-20240923/TWL_Data_For_Massey_Protected.xlsx';

PROC IMPORT DATAFILE=REFFILE
    DBMS=XLSX
    OUT=WORK.TWL_DATA
    REPLACE;
    GETNAMES=YES;
RUN;
data work.TWL_DATA_CLEAN_DUMMY;
    set work.TWL_DATA_LABEL;
```

[Data Processing]

```
    /* Delete sales less than 0.5 */
    if TotalSales > 0.5;

    /* Initialize all dummy variables to 0 */
    Holiday_Dummy_2017 = 0;
    Holiday_Dummy_2018 = 0;
    Holiday_Dummy_2019 = 0;

    /* Set dummy for the corresponding peak periods */
    if '19DEC2017'd <= CalendarDate <= '26DEC2017'd then Holiday_Dummy_2017 = 1;
    else if '19DEC2018'd <= CalendarDate <= '26DEC2018'd then Holiday_Dummy_2018 = 1;
    else if '19DEC2019'd <= CalendarDate <= '26DEC2019'd then Holiday_Dummy_2019 = 1;

    /* Delete extreme values        greater than 4.5 at other times */
```

```
    if (TotalSales <= 4.5) or (Holiday_Dummy_2017 = 1 or Holiday_Dummy_2018 = 1 or
Holiday_Dummy_2019 = 1);
run;
```

[Calculate Adstock Variables]

```
/*code to calculate ad stock for advert sales data*/
data work.Advert_Sales_AdStock;
  set work.TWL_DATA_SORT;

lambda_t = 0.9;
lambda_r = 0.9;
lambda_d = 0.9;
lambda_m = 0.9;
lambda_p = 0.9;

  /* Initialize the adstock variables */
if _N_ = 1 then do;
television_adstock = LN_Television;
radio_adstock = LN_Radio;
digital_adstock = LN_Digital;
mailer_adstock = LN_Mailer;
press_adstock = LN_Press;
retain television_adstock radio_adstock digital_adstock mailer_adstock press_adstock;
end;

/* Calculate adstock based on lambda values */
else do;
television_adstock = sum(lambda_t * television_adstock, (1 - lambda_t) * LN_Television);
radio_adstock = sum(lambda_r * radio_adstock, (1 - lambda_r) * LN_Radio);
digital_adstock = sum(lambda_d * digital_adstock, (1 - lambda_d) * LN_Digital);
```

```
mailer_adstock = sum(lambda_m * mailer_adstock, (1 - lambda_m) * LN_Mailer);
press_adstock = sum(lambda_p * press_adstock, (1 - lambda_p) * LN_Press);
retain television_adstock radio_adstock digital_adstock mailer_adstock press_adstock;
end;

format television_adstock radio_adstock digital_adstock mailer_adstock press_adstock 8.2;
run;
```

References

Doe, J., & Smith, A. (2020). Multi-channel advertising effects on consumer behavior. Journal of Marketing Research, 57(3), 451-472.

Johnson, M., & Brown, L. (2019). Optimizing ad spend across TV and digital platforms. International Journal of Advertising, 38(4), 525-540.

Lee, K., & Wang, Y. (2018). The impact of adstock decay rates on long-term brand equity. Marketing Science, 37(1), 102-117.

Patel, R., & Kim, S. (2021). Machine learning in advertising: Automated budget allocation for maximizing ROI. AI & Marketing Journal, 45(2), 87-99.